

a light, simple and cheap part with adequate strength can be produced.

Now if these conditions have not been met, the effect which the manufacturing facility has on the material properties, such as strength, should be manifested and hence contribute towards the failure of the machine elements / components.

Frequent failure of components due to lack of qualitative parts produced and also lack of adequate strength to withstand the working load would be experienced.

At this juncture, the failure analysis of the components can be carried out to determine the mode and mechanism, leading to their failure. After which the ways to prevent their future occurrence can be suggested.

Types of Joints

To know the use of mechanical fasteners and joints in making common products is essential, and so are the aspects of the design method applicable to the typical joints held in place by rivets, bolts, screws, weld seams, and adhesive materials, etc. The basic function of a mechanical joint is to transmit a given load from one piece of material to the adjacent one, thereby reducing stress concentration in joints.

Joints are required to connect two or more machine elements. The joints may be made for the parts to be connected to gether permanently or disconnected for repairs or replacement.

Hence joints can be classified as :

2) Non-Permanent Joints; Otherwise, ~~se~~ known as temporary or detachable joints, are joints that can be disconnected without destroying the joints. In other words, these are the joints which can be parally disassembled into separate elements without damaging or destroying them. Bolted joints, screws joints, key or keyed joint, coupling joint, cotter joint, knuckle joint, etc., are their examples.

b) Permanent joints: are joints that can be disconnected only ~~at~~ ^{under} ~~in~~ ^{special} ~~circumstances~~ ^{conditions}. Welded joint, welded joint, etc, are their exam-

Mechanics itself can be made by using first years and other methods.



(1)

Fasteners include bolts and nuts, rivets, screws, cotter's, locking pins, etc.

RIVETS AND THEIR TYPES

A rivet is a short cylindrical bar with a head at one end, tail on the other end and shank or body in between. In other words, it is a fastening device which is secured by distortion or upsetting of the shank and ends. The rivets are used to connect together permanently two or more plates or other pieces of metal. They are machine forged with a single head. When used the head is held in place and a second head is formed from the projecting shank, either by hand or by machine. Hence, riveting is done by pressing the head of the rivet with the help of hydraulic pressure, pneumatic pressure ~~or~~ ^{so it is an operation whereby two plates are joined with the help of a rivet} or hammer.

In hand riveting, the original head is backed up by a hammer or heavy bar. Then a die is placed against the end to be headed and the blows are applied by the hammer. This forms the second head called the point. Whereas in machine riveting, the die is part of the machine which is operated by air, hydraulic or steam pressure. Rivets are closed red hot under hydraulic pressure, because the steady pressure produced on the end bar of the rivet stays it up better and causes it to fill the hole more completely than is likely to be the case with steam, pneumatic or hand riveting. Hence the action is percussive.

(12)

Cold riveting is used when the steel rivets are up to 12mm in diameter. For larger diameters, hot riveting is used where the entire length of the rivet is heated to bright red.

Rivets are available in different materials. They are made of wrought iron or soft steel for most cases, but where corrosion resistance or light weight is a requirement, rivets of copper or aluminium alloys are used.

They are widely used in structural works, ship building, bridges, tanks and pressure vessels.

Rivet heads are of different types:
~~Some of which are:~~

(i) Snap head: is used for structural work and machine riveting.

(ii) Conical head: is mainly used where restricted space only permits of head hammering.

(iii) Counter sunk head: is used in ship building where flush surfaces are necessary.

Joining of sheets or plates of any metal by means of rivets is called a riveted joint. Based on the arrangement of plates or their like which need to be connected, the riveted joints are classified as:

A) Lap joint: A lap joint is that in which one plate overlaps the other and the two plates are then riveted together.

Based on the number of lines / rows of rivets used, the lap riveted joint is further classified as:

1. Single riveted lap joint: is that in which there is a single row of rivets on

the lap joint. Refer to fig 1 below.

2. Double riveted lap joint: In this in which there are two rows of rivets on the lap joint. Refer to fig 2 below.

3. Triple riveted lap joint: In this in which there are three rows of rivets on the lap joint. Refer to fig 3 below.

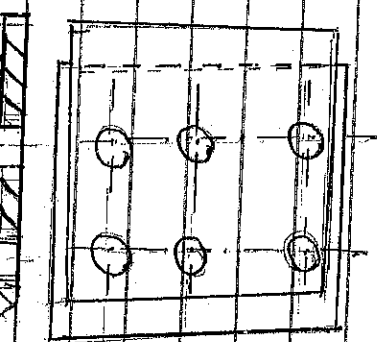
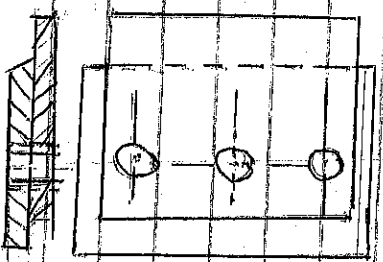


Fig. 1 Single riveted lap joint

Fig. 2. Double riveted lap joint / Double riveted chain type lap joint

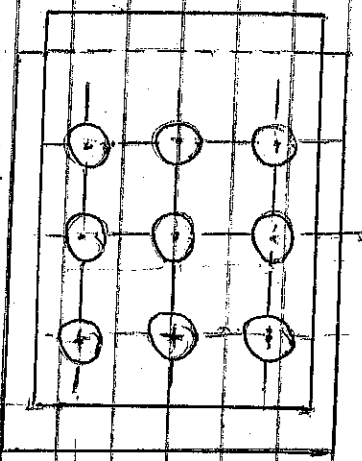


Fig. 3 Triple riveted lap joint / Triple riveted chain type lap joint

Based on the arrangement of rivets in the various rows as being staggered or in line, the double and triple lap joints are further classified as:

a) Double riveted chain type lap joint

is that in which there are two rows of rivets, which are arranged opposite to each other on the lap joint. Refer to fig. 2 in the above.

b) Triple riveted chain type lap joint: is that in which there are three rows of rivets, which are arranged opposite to each other on the lap joint. Refer to fig. 3 in the above.

Also, based on the arrangement of rivets in the adjacent rows a being 'zigzag' or 'chain' the double and triple riveted lap joints are further classified as:

(i) Double riveted zigzag type lap joint: is that in which there are two rows of rivets, which are staggered in such a way every rivet is in the middle of two rivets of the opposite row in the lap joint. Refer to fig. 4 below.

(ii) Triple riveted zigzag type lap joint: is that in which there are three rows of rivets, which are staggered in such a way every rivet is in the middle of two rivets of the opposite row in the lap joint. Refer to fig. 5 below.

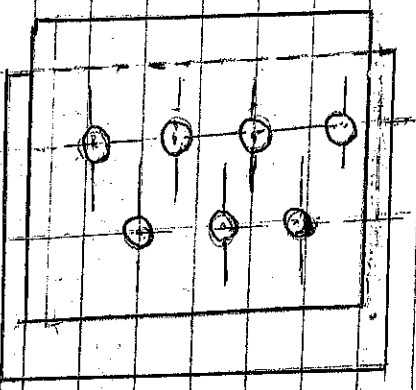


Fig. 4

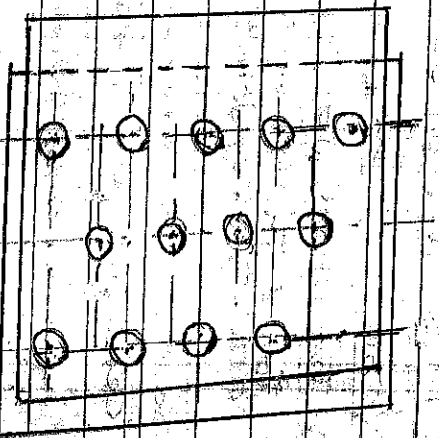


Fig. 5

B) Butt joint: A butt joint is that in which the main plates are kept in alignment butting (touching) each other and a cover plate (or plates) each other and a cover plate is affixed across a strap is placed on either one side or both sides of the main plates. The cover plates are then riveted with the main plates. The thickness of the cover plate is usually less than that of the main plate.

- Based on the number of cover plates used, the butt riveted joints are classified as:

1. Single strap butt joint: In butt joints, in the butt joint, if only one cover plate is placed on one side of the main plates, then such a butt joint is called a single strap butt joint.

3. Double strap butt joint: In butt joint, in the butt joint, if two cover plates are placed on both sides of the main plates, then such a butt joint is called a double strap butt joint.

Based on the number of lines/rows of rivets used, the single strap butt joints, are further classified as:

- Single riveted single strap butt joint
- Double riveted single strap butt joint
- Triple riveted single strap butt joint
- and quadruple riveted single strap butt joint.

In such butt joints, although the edges of the main plates after butt against each other and only one cover plate is placed on one side of the plates and then riveted together, there could be one, two, three or quadruple rows of rivets passing through each of the plates depending on whether it is a single/double/triple and quadruple riveted joints. Refer to figures

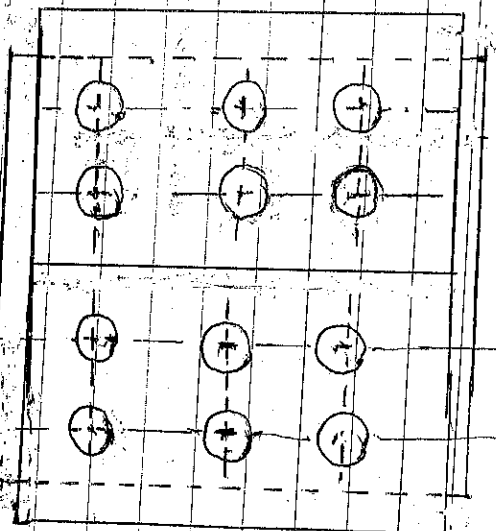
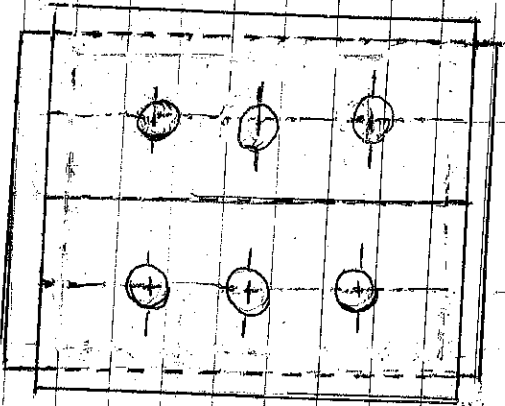


Fig. 6 Single riveted strap butt joint



Fig. 7 Double riveted strap butt joint

Double riveted strap butt joint. Double riveted single strap chain type butt joint.

And these of the double strap are further classified as: single riveted double strap butt joint, double riveted double strap butt joint, triple riveted double strap butt joint and quadruple riveted double strap butt joint.

In such butt joints, ~~the plates are placed against each other~~ plates butt against each other and two cover plates are placed on both sides of the plates and then riveted together, there ~~shall be one, two or three or four rows of rivets passing through each of the plates depending on whether it is a single, double, triple or quadruple riveted joint.~~ Refer to figures 8 and 9 below,

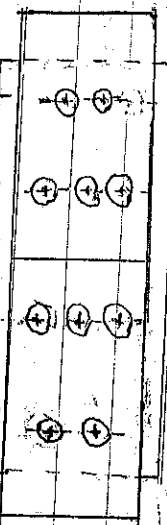


Fig. 8 Single riveted strap zigzag butt joint

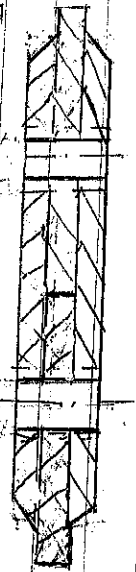
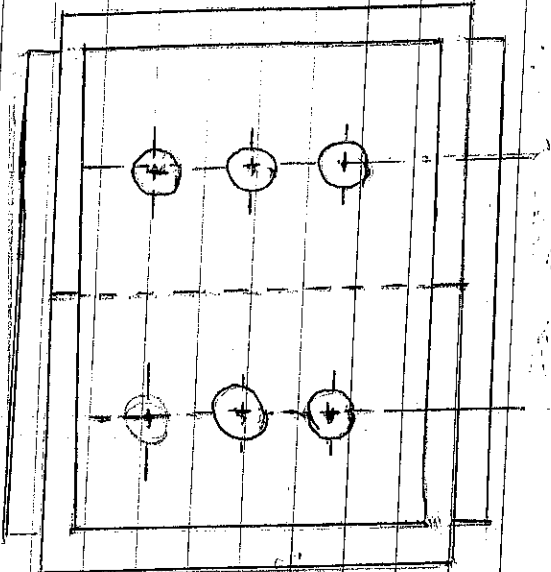


Fig. 8. Single riveted double strap butt joint

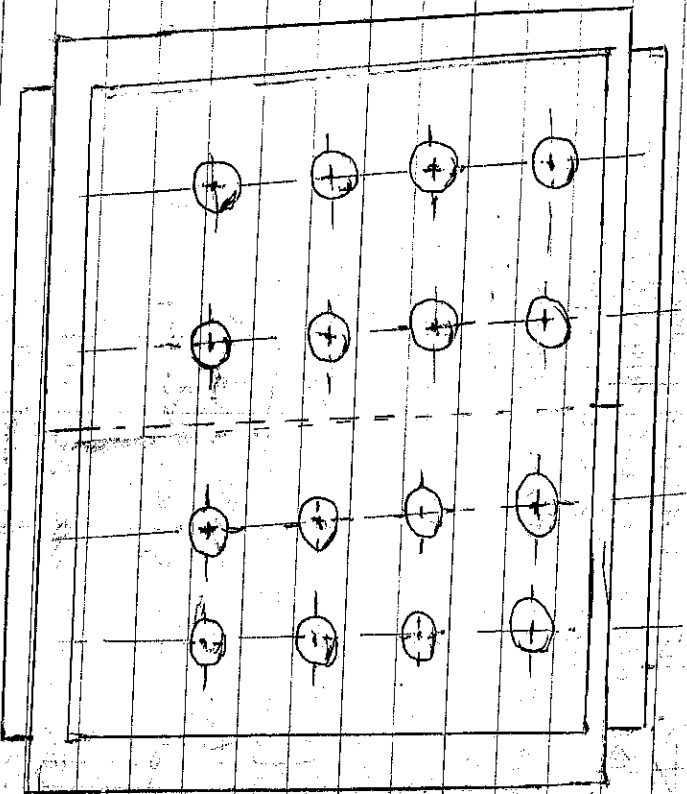


Fig. 9. Double riveted double strap butt joint
Double riveted double strap chain type butt joint

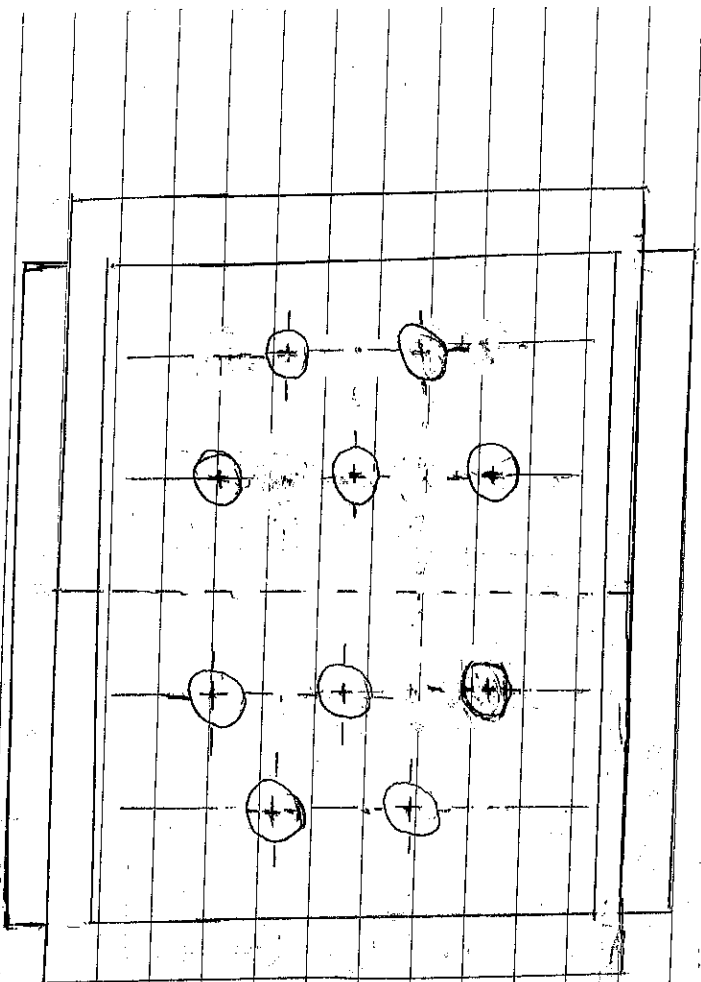


Fig. 10. Double riveted double strap zig-zag type butt joint

Based on the arrangement of rivets as they are opposite to each other ~~the double strap~~ higher riveted single strap butt joints are further classified as:

Double riveted single strap chain type butt joint, triple riveted single strap chain type butt joint and quadruple riveted single strap chain type butt joint.

In such butt joints, although only one cover plate is ~~placed~~ on one side of the main plates, the rivets in the ~~main plates~~ rivets in each plate are opposite to each other in same transverse line. Refer to fig. 11(a) & (b). Likewise, those of the double and higher riveted double strap are further classified as:

Double riveted double strap chain type butt joint, triple riveted double strap chain type butt joint and quadruple riveted double strap chain type butt joint.

In such butt joints, the two cover plates need are riveted together with the main plates and there could be two, three and four rows of rivets passing through each of the plates depending on whether it is a double, triple and quadruple riveted joint. However in this case, the rivets in one row and those of the next are opposite to each other or in ~~some~~ transverse line. ~~Ropes to fig. 19 above~~ Similarly brace on the arrangement of rivets in zig-zag form, the double and higher riveted single strap butt joint are further classified as:

Double riveted single strap zig zag type butt joint, triple riveted single strap zig zag type butt joint and quadruple riveted single strap zig zag type butt joint.

In such butt joints, although the single strap need is riveted together with the main plates, the two, three and four rows of rivets passing through each of the main plates are not opposite to each other nor in same transverse line, but rather the rivets in the adjacent rows are staggered in such a way that every rivet is in the middle of the two rivets of the opposite row. Refer to fig. 19b ~~to the above~~.

And those of the double and higher riveted double strap, are further classifying as:

(20)

Double riveted double strap zig-zag type butt joint, triple riveted double strap zig-zag type butt joint and quadruple riveted double strap zig-zag type butt joint.

In such butt joints, although the two cover plate bevel on both sides of the main plates are riveted together with the main plates, the two, three and four rows of rivets passing through each of the plates are staggered in such a way that every rivet is in the middle of the two rivets of the opposite row. Refer to fig. 10 in the above.

CLASSIFICATION OF RIVETED JOINTS IN SUMMARY

LJ = Lap joint

BJ = Butt joint

S = Single

SR = Single riveted

SS = Single strap

D = Double

DR = Double riveted

DS = Double strap

T = Triple

TR = Triple riveted

Q = Quadruple

QR = Quadruple riveted

C = Chain

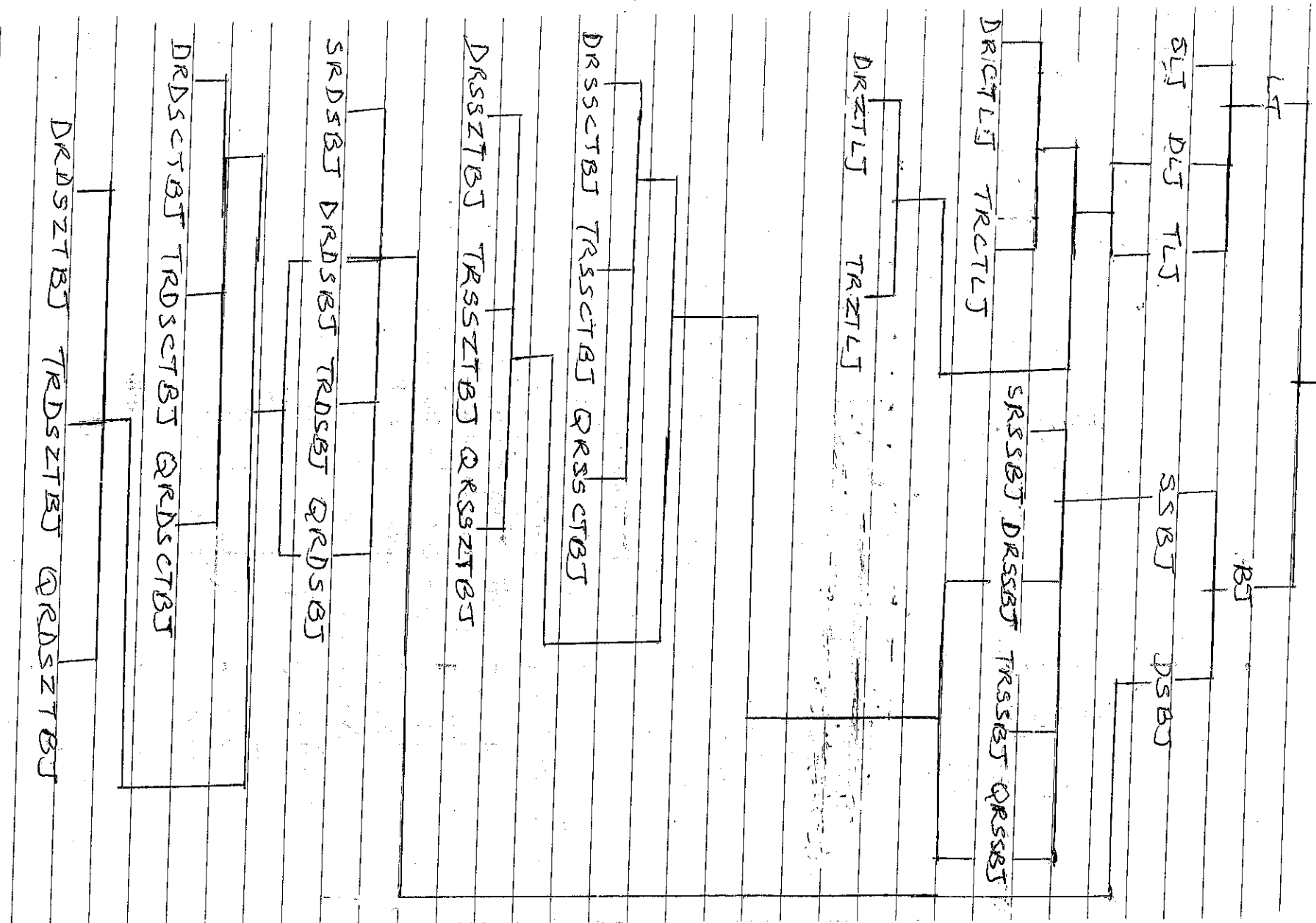
CT = Chain

Z = Zig-zag

ZT = Zig-zag type

(21)

Riveted joints



(22)

TERMS RELATED TO RIVETED JOINT

Consider a double riveted lap or a double riveted unequal double strap zigzag type butt joint as shown.

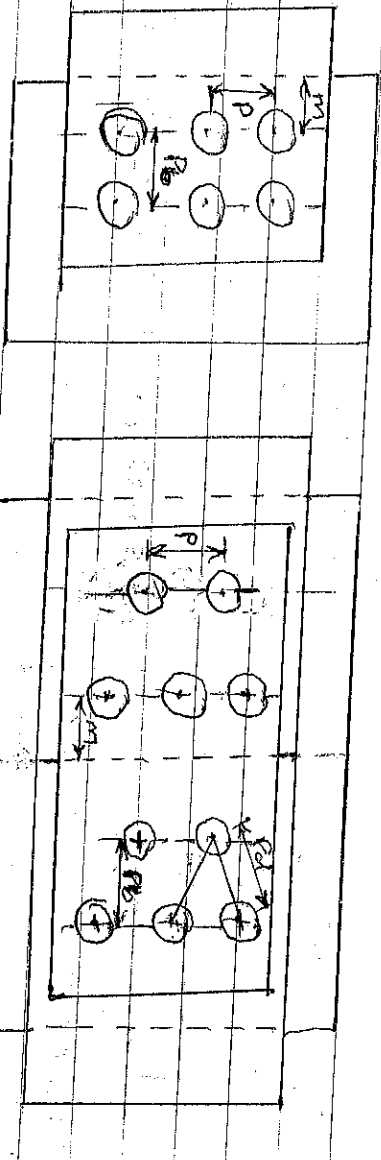


Fig. 11(a) Double riveted lap joint

Fig. 11(b) Double riveted unequal strap zigzag type butt joint

Pitch: is the distance from the centre of one rivet to the centre of the next rivet measured parallel to the seam. It is denoted by P .

Back Pitch: is the perpendicular distance between the centre lines of the successive rows. It is denoted by P_0 .

Diagonal Pitch: is the distance between the centres of the rivets in the adjacent rows of zigzag riveted joint. It is denoted by P_d .

Margin: Sometimes called marginal distance or pitch, is the distance between the centre of rivet hole to the nearest edge of the plate. It is denoted by m .

DESIGN OF RIVETED JOINTS

The usual methods employed in the design of joints are based on the apparent direct stress, with the unknown stress provided for in the factor of safety. The assumption for the design of riveted joints are:

- a) The load is distributed among the rivets according to the shear area.
- b) There is no bending stress in the rivets.
- c) The tensile stress is equally distributed over section of the member between the rivets.
- d) In a rivet subjected to double shear, the shear is equally distributed between two areas in shear.
- e) The holes into which the rivets are driven do not weaken the member if it is in compression.
- f) After driving, the rivets completely fill the holes.

(g) Friction between adjacent surfaces also does not affect the strength of the joint.

In designing riveted joints of pressure vessels, the diameter of the hole of the rivets both in shear and crushing. In structural joints, the rivets sometimes are driven cold and sometimes are replaced by bolts, it is therefore customary to use the nominal rivet diameter in calculating the resistance of rivets.

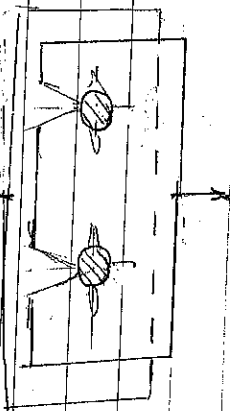
The strength of a joint is evaluated

(24)

by tearing are possible failure paths in the joint into account and it is usually ~~not~~ at least by considering one pitch length of each joint.

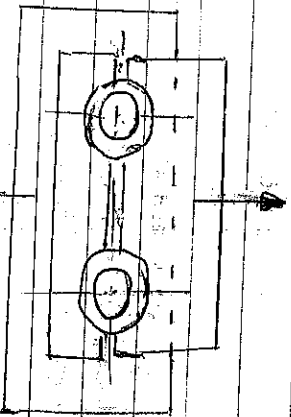
The different ways in which the joint may fail are:

1. The plate may tear at the front of the rivet as shown.



This may be due to the bending of the plate. To avoid this, the edge distance of the rivet should be $m = 1.5 d_o$, which will be safe against both shearing and tearing by the rivet pressure. For the sake of additional safety, in single rivet joint, $m = 2 d_o$. Where d_o = Diameter of rivet hole.

2. The plate may tear through the centre of the holes as shown.



This may be due to tensile stresses set up in the main plate or the plate length between the holes along the pitch line is considered to be the weak link.

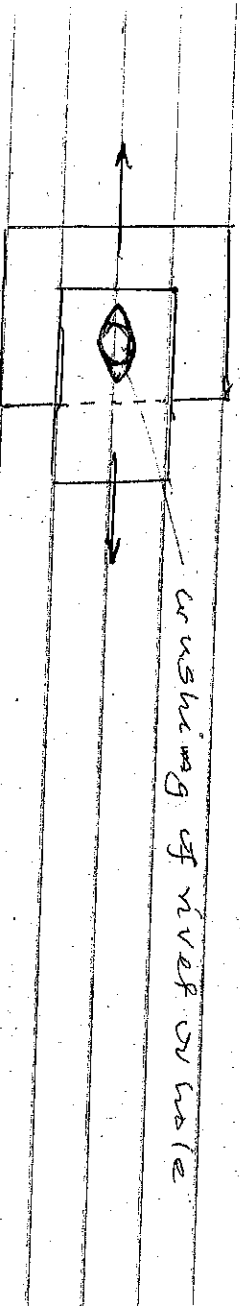
Plate resistance to tension, F_t

= Area of plate along line $\times$ Allowable tensile stress = $(p - d_o) t \cdot \sigma_t$ — (2)

(25)

Where P = Pitch of the rivets, mm, d_o = Diameter of the rivet hole, mm, t = Thickness of the plate, mm, σ_t = Allowable tensile stress of the plate material, N/mm^2 .

3. Plate and rivets may crush or fail in compression as shown.



This makes the joint loose.

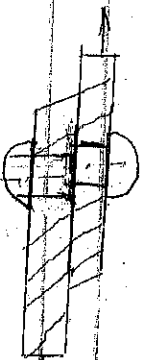
The resistance to crushing, F_c
 $= d_o t \sigma_c$ — (3)

Where d_o = Diameter of the rivet hole, mm,
 t = Thickness of the plate, mm.

σ_c = Allowable crushing stress between the rivet and plate material, N/mm^2 .

Z = Number of rivets in a pitch length, dimensionless.

4. The rivets may shear as shown.



a) Single shear,

b) Double shear,

The resistance to shear, F_s
 $= \bar{n} d_o \tau$ — (3)

Where Z = Number of rivets per pitch length, mm,
 d_o = Diameter of the rivet hole, mm, τ = Allowable shear stress of the rivet material, N/mm^2 .

$\bar{n}$ = Number of shears, dimensionless.

= 1 for single shear and 2 for double shear.
= 1.875 for double shear on perpendicular boiler

connections (I.B.R.).

(26) for

It is worth noting that the lap joints and butt joints with single strap, the rivets will be in single shear. And for butt joints with two straps, the rivets will be in double shear. Also the above expressions (2) and (3) are for single riveted lap joints in which $z = 1$. While expression (4) is applicable to all types of joints as it is per pitch length, and so is the strength of the plate expression. Now double and triple riveted joints may fail by a combination of shearing, crushing and tearing.

Consider the double riveted zig-zag type lap joint shown in the fig. 4 above (i.e. page 14).

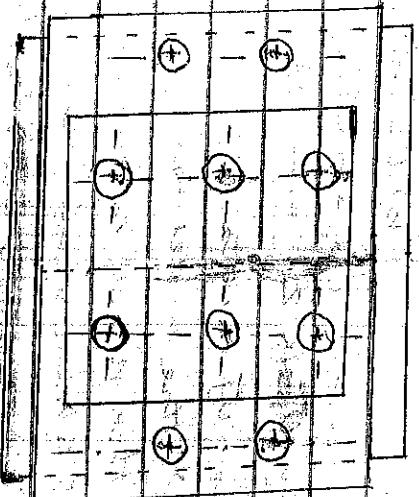
Resistance to tearing between two rivets, $F_t = (p - d) \sigma_t$

Resistance to crushing in front of two rivets, $F_c = 2 d \sigma_c$ (Since $z = 2$)

$$Resistance to shearing two rivets, F_s = 2 \left(\frac{\pi}{4} d^2 \right) \tau_s \quad (\text{Since } z = 2 \text{ and } k = 1)$$

Strength of solid plate, $F_p = P \sigma_t$

Consider a double riveted zig-zag type lap joint should strap zig-zag type type butt joint - as shown



Resistance to tearing at outer end of rivets, $F_t = (p - d) \sigma_t$



(27)

Resistance to shearing one rivet in double shear and one in single shear

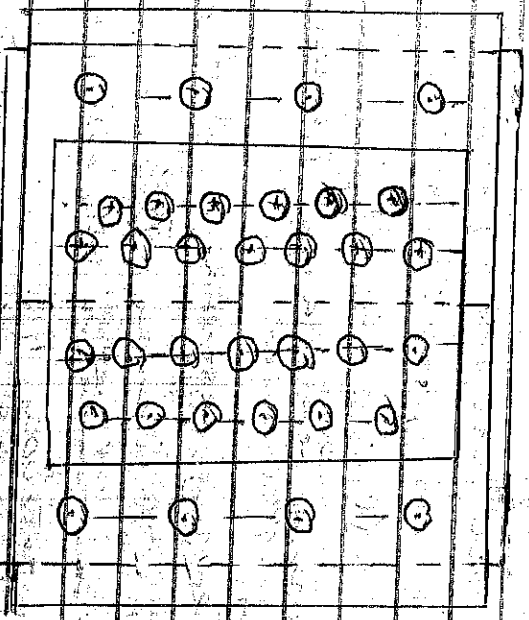
$$F_s = 1.5 \left(\frac{\pi d_o}{4} \right)^2 \tau_s + \frac{\pi d_o^2}{4} \tau_s$$

Resistance to tearing at inner row of rivets and shearing one of the outer row of rivets $F_{t+s} = (p - d_o) t + \frac{\pi d_o}{4} \tau_s$ since $k=1$
 two rivets $= 2 d_o t + 2 d_o t \tau_s$

Combining in front of one rivet and shearing one rivet $F_t + F_s = d_o t + \frac{\pi d_o^2}{4} \tau_s$

Strength of solid plate $= P t$

Consider a triple riveted butt joint as shown. Cover plate zig-zag type



Resistance to tearing at outer row of rivets, $F_t = (p - d_o) t$

Resistance to shearing four rivets in double shear and one in single shear, $F_s = 4 k \left(\frac{\pi d_o^2}{4} \right) \tau_s + \frac{\pi d_o^2}{4} \tau_s$

Resistance to tearing at middle row

(28)

of rivets and shearing one rivet,
 $F_t + F_s = (P - 2d_0) t + 0.7 + \pi d_0^2 \tau_s$

4

Resistance to crushing in front of four rivets and shearing one rivet,
 $F_c + F_s = 4 d_0 t \sigma_c + \frac{\pi d_0^2 \tau_s}{4}$

Resistance to crushing in front of five rivets, $F_c = 5 d_0 t \sigma_c$

Strength of solid plate, $F_p = P t \sigma_t$

EFFICIENCY OF RIVETED JOINT

The efficiency of a riveted joint may be defined as the ratio of the least that will

produce the allowable tension stress in the unpunctured plate. It is usually represented as percentage.

Tearing efficiency, η

Tearing resistance of weakest part of strip = $(P - d_0) t \sigma_t$
Tearing resistance of unpunctured plate = $P t \sigma_t$

Shearing efficiency, η_s

Resistance to shearing of rivets = $\frac{A d_0 \tau_s}{4}$
- of rivets in a strip
Tearing resistance of unpunctured plate = $P t \sigma_t$

Crushing efficiency, η_c

Resistance to crushing of rivets or holes in a strip = $d_0 t \sigma_c$
Tearing resistance of unpunctured plate = $P t \sigma_t$

The lowest of the three efficiencies is the efficiency of the joint.

Or The ratio of the strength of the riveted joint to the strength of the unpunctured plate per pitch length is called efficiency of the riveted joint. Mathematically, the efficiency of three rivets

(29)

ed joint,

$$\eta = \text{least value of } \frac{P_t}{P_t + P_s} \text{ and } \frac{P_t}{P_t + P_s}$$

Usually, the strength of the riveted joint is equal to the least value of P_t , P_s and P_e .

DESIGN OF BOILER JOINT

In general, for longitudinal joint, butt joint is adopted, while for circumferential joint, lap joint is preferred. Hence these are the two types of joints needed to fabricate a boiler.

DESIGN OF LONGITUDINAL JOINT

The longitudinal butt joint is usually a double strap butt joint that is used to form a cylindrical shell by joining the ends of a plate. It is more efficient than lap joint. It is stiffer also and helps to maintain circularity of the shells at the ends of the plate. Its joints are staggered because the shell is made up of a number of rings which are connected at plates.

1. Thickness of the plate: The thickness of the boiler shell is determined by using the cylinder formula, that is,

$$t = \frac{P D}{2 \sigma \eta} + 1 \text{ mm (Corrosion allowance)}.$$

Where P = Internal pressure, N/mm^2 , D = internal diameter of the boiler shell, N/mm^2 .

σ = Permissible stress of the shell material, N/mm^2 , η = Efficiency of the joint.

2. Diameter of the nozzle: It is given by various empirical formulae, that is,

$\delta_0 = 6.05\sqrt{t}$ for $t \leq 8 \text{ mm}$.

Where $t = \text{Thickness of the boiler shell mm}$.

But if the plate thickness is less than

8mm, the diameter of the rivet hole is determined by equating the shearing resistance of the rivet to its crushing resistance and taking corresponding value of the stress of rivet diameter in table 9.2. If the rivet diameter may be obtained by

equating the tearing resistance of the plate to the shearing resistance of the rivets. Hence its minimum value should not be less than $2\delta_0$ and its maximum value for a longitudinal joint is given by

$$P_{\text{max}} = C t + 41.28 \text{ mm}$$

Where C is a constant depending on the type of joint.

However, if the pitch of the rivets is obtained by equating the tearing resistance to the shearing resistance is more than

max then the value of P_{max} can be adopted. And the value for pressure vessel is 3.0 for DRG.

4. Row (transverse) pitch / bore pitch.

- (i) For equal number of rivets in more than one row for lap joint and butt joint the row pitch should not be less than $(0.33P + 0.67\delta_0)$ for zigzag riveting and $2\delta_0$ for chain

riveting. All these expressions and its value are by

(ii) For joints in which the number of rivets

in the row is half the number of rivets

in the inner rows and if the inner rows are chain riveted then the distance between the outer row and the next row

should not be less than $(0.33P + 0.67\delta_0)$ or $2\delta_0$ whichever is greater. The

distance between the rows in which

there are full number of rivets should

not be less than $2\delta_0$.

(31)

(iii) For joints in which the number of rivets in the outer row is half the number of rivets in inner rows, and if the inner rows are chain riveted, the distance between the outer rows and the next row, should not be less than, $0.25P + 1.5d_o$. The distance between the rows in which there are full number of rivets (2nd-3rd) should not be less than $0.165P + 0.67d_o$.

Note: P is the pitch of rivets in the outer row & d_o is the thickness of butt straps. Its

their thickness is given by (in no case), it should not be less than $t_1 = 1.125t$, for ordinary single-butt strap (chain riveting).
butte strap (chain riveting).

OR $t_1 = 1.125t \left[\frac{P-d_o}{P-2d_o} \right]$ for a single butt strap with alternate rivets in the outer rows omitted.

OR $t_1 = 0.625t$, for ordinary double straps of equal width (chain riveting).

OR $t_1 = 0.625t \left[\frac{P-d_o}{P-2d_o} \right]$ for double straps of equal width, with alternate rivets in the outer rows omitted.

When two unequal width of butt straps are employed, the thickness of butt straps are given by,

$t_1 = 0.75t$, for wide strap on the inside, and

$t_2 = 0.625t$, for narrow strap on the outside.

Where $t =$ Butt strap thickness and $m =$ The number of rivets in the outer row.

7. The elongation pitch (in case of

Zig-zag joint) (32)

$$P_d \geq (P + d_o) / 2$$

$$\text{OR } P_d = (2.25 \text{ to } 2.50) d_o.$$

Note: Centre from (C) above, the distance between rivets, P_d , should not be less than 0.6 times pitch for zigzag riveting and 0.8 times pitch for chain riveting.

DESIGN OF CIRCUMFERENTIAL LAP JOINT

The circumferential lap joint is one made with one ring overlapping the other alternately.

1. The thickness of the boiler shell, the diameter of the rivet hole and the diameter of the rivets are adopted as the same as those used for longitudinal joint.

2. Number of rivets: The rivets are in single shear, since lap joint is used for circumferential joint. The total number of rivets to be used for the joint is given by

$$Z = \text{Steam load}$$

Shear strength of one rivet

$$= \frac{\pi D^3 P_s}{4 \sqrt{d_o} T_s} = \left(\frac{D}{d_o} \right)^2 \cdot \frac{P_s}{T_s}$$

Where D = Inner diameter of the boiler, mm.

P_s = Steam pressure, N/mm^2 .

d_o = Diameter of the rivet hole, mm.

T_s = Allowable shear strength of the

rivet material, N/mm^2 .

Or Z = Steam load

Crushing strength of rivets in the steam

$$= \frac{\pi D^2 P}{4} \quad \text{Where } D = \text{Allowable crushing stress between the rivet and plate material.}$$

The greater of the two values of Z that are found is taken as the basis of the Seam Design. If the load coming on the is of varying pipe, and varies between $F_{\max}$ and $F_{\min}$, the allowable stress for a work made from a Carbon Steel is decreased by multiplying it by the correction factor

$$1 - \frac{F_{\min}}{3F_{\max}} \leq 1. \quad \text{The force } F_{\min} \text{ and } F_{\max} \text{ appear in the formula each with its sign.}$$

3. Pitch of the rivets. In general, the of the circumferential lap joint is one half of the efficiency of the longitudinal joint. Hence once the efficiency of the longitudinal joint is known, the efficiency of the circumferential joint can be calculated. At times, the magnitude of the circumferential joint efficiency may be ~~known~~ ^{known} that depending on how large is the pitch of the rivet or the number of failure considered or the strength of the joint.

Assuming this to be the lowest efficiency of the circumferential lap joint in tearing, the pitch of the rivets can be calculated from relation

$$p = P - d_0$$

4. Number of rivets: is determined thus, the number of rivets per row has

(34)

to be determined, that is

$$Z = \pi (D + t)$$

P

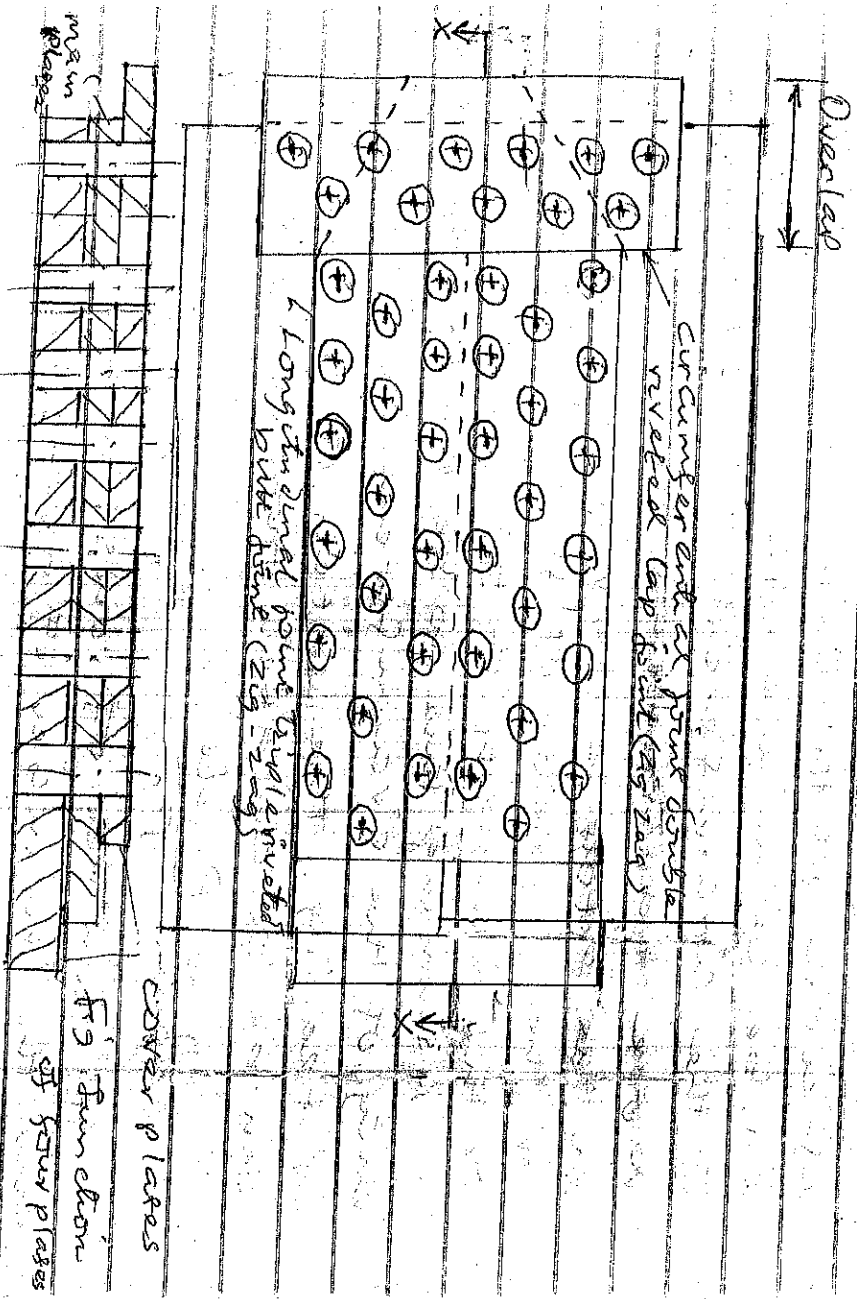
And the number of rows is given by

$$n = \frac{\text{Total number of rivets}}{\text{No. of rivets per row}}$$

5. Selection of the type of joint: After determining the number of rows, the type of joint, such as single rivet, double rivet, etc, may be decided.

6. Rivet heads, pitch is calculated as mentioned in the above, and so is the margin.

7. When the longitudinal joint is in the circumferential joint on steam boiler, the end of the longitudinal strap fit down underneath the circumferential plate in a recess to receive it as shown.



(35)

When the distance between the nose of rivets is known, the overlap of the plate can be fixed.

$$\text{Overlap} = (\text{Number of rows} - 1) P_b + 2m.$$

THREADED FASTENERS

Bolts, screws and studs are the most common types of threaded fasteners. They are used in both permanent and removable joints.

Bolts: are basically threaded fasteners normally used with nuts.

Screws: engage either a preformed or a self-made internal thread.

Studs: are externally threaded headless fasteners. One end meets with a ~~thread~~ ^{nut} and the other with standard nut.

There are different forms of bolts and screws heads for different usage. These include bolt heads of square, hexagonal or eye shape and screw heads of hexagonal, filister, button head, counter sunk or Phillips type. These are as shown.

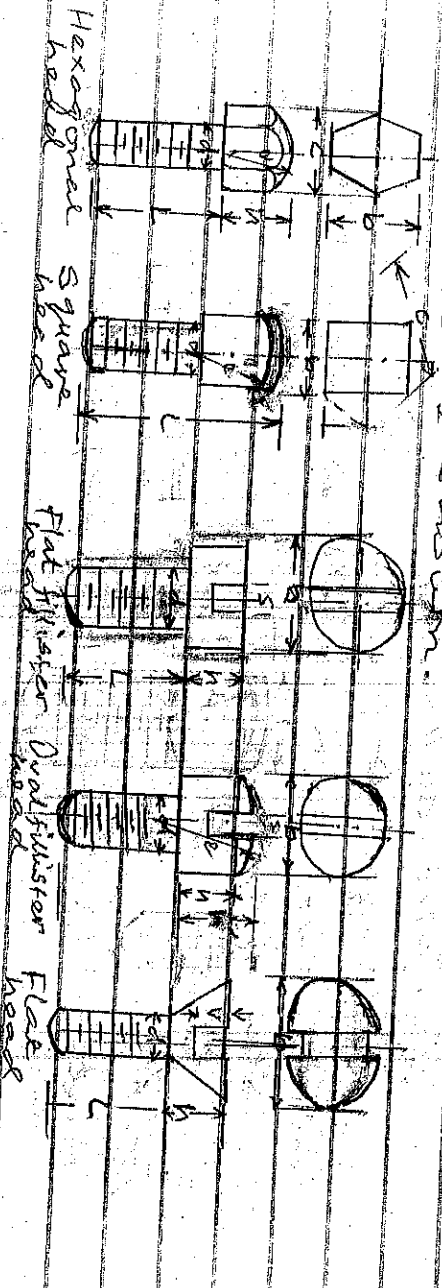


Fig. Types of bolt heads

(36)

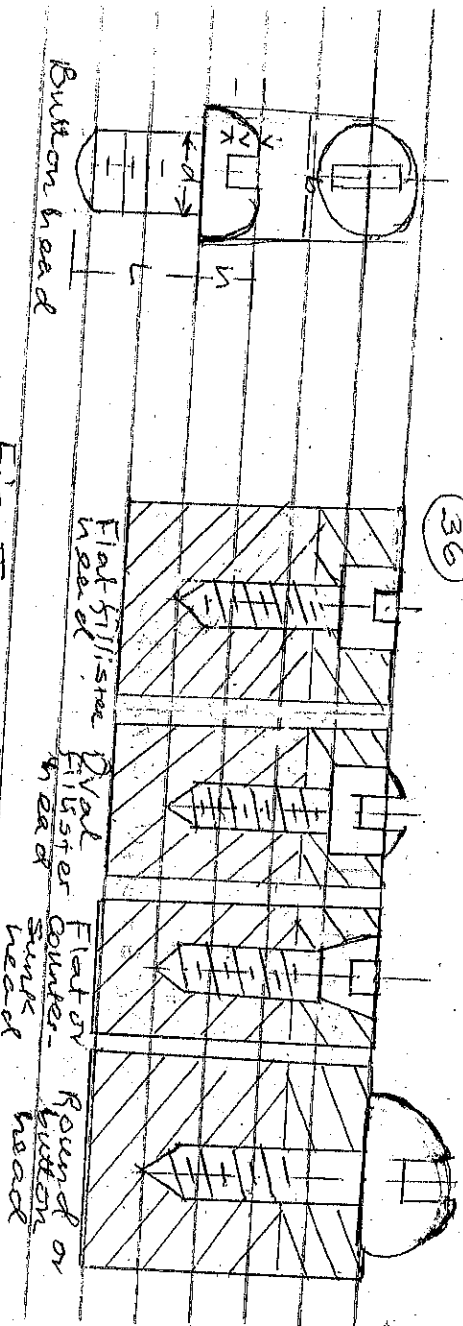


Fig. Types of Screw heads.

TAPPING SCREWS

These are one piece fasteners which cut or form a mating thread when driven into a preformed hole. These allow rapid installation since nuts are not used.

There are two types of tapping screws. They are known as threaded forming which displaces or forms the adjacent materials and thread cutting which have cutting edges and chip cavities which create a mating head.

SET SCREWS

These are semi-permanent fasteners which hold collars, pulleys, gears, etc, on a shaft. Different heads and point styles are available. ~~Some of the~~ and

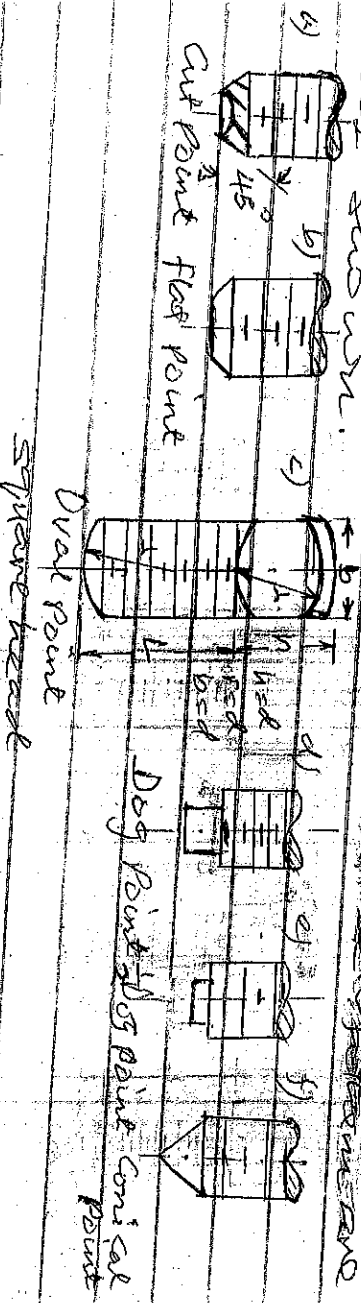
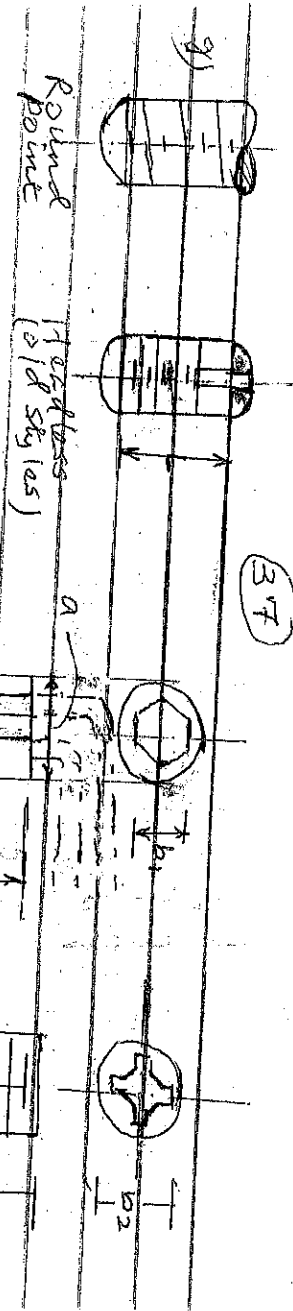


Fig. Different types of Set Screws

(317)



Allen headless

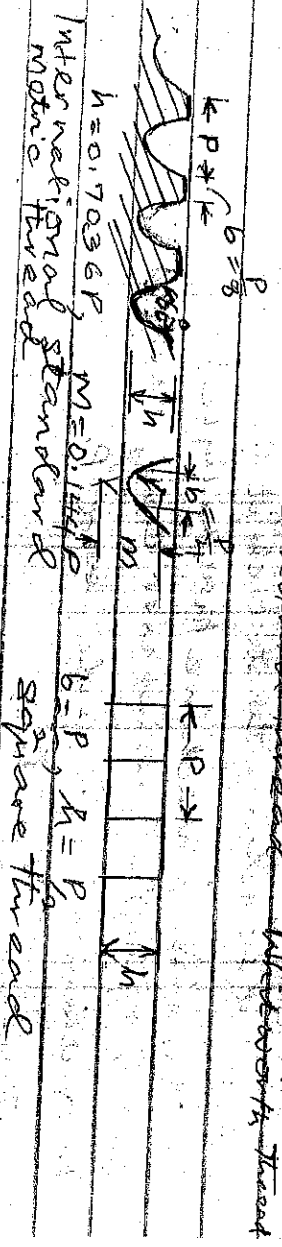
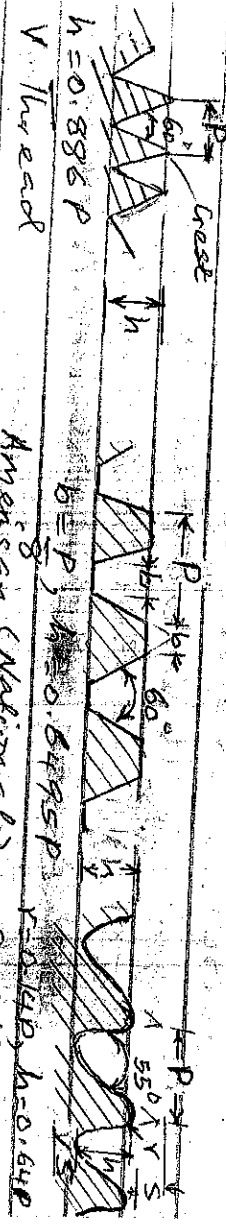
Bristo headless



THREAD FORMS

Basically when a helical groove is cut or generated over a cylindrical or conical section, threads are formed. When a point moves parallel to the axis of a rotating cylinder or cone held between centres, a helix is generated. Screws forms in this way have two functions to perform in general.

Namely: (i) To transmit power by square, Acme, Buttress, knuckle types of threads, (ii) To secure one member to another by V-threads or studs.



(38)

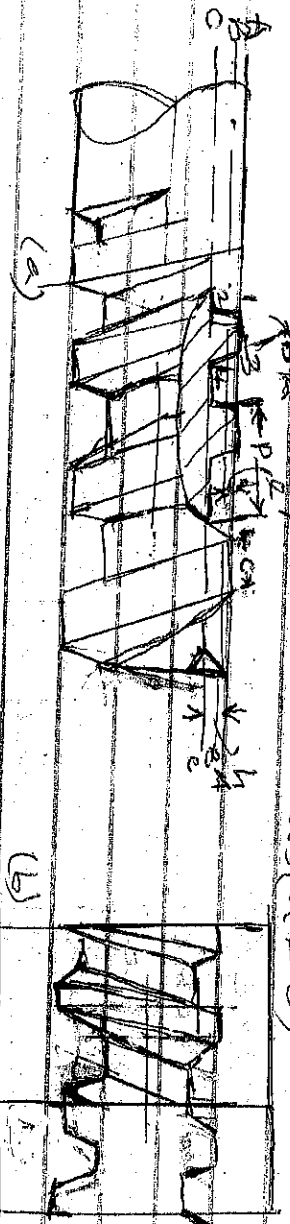
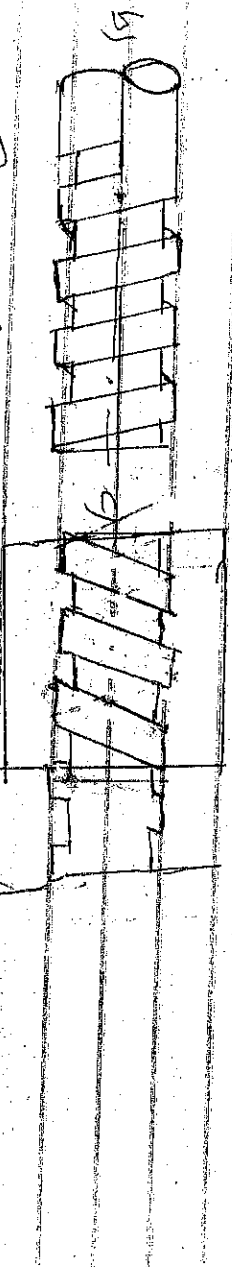
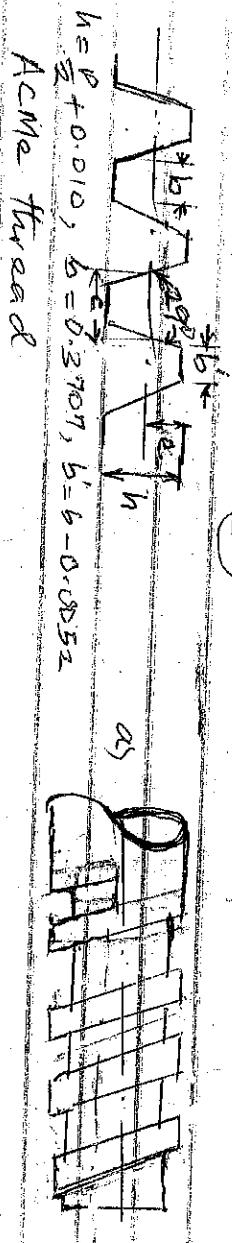


Fig. Different types of thread forms

Terms referred to threads such as minor / core / root diameter, major / outside diameter, mean / effective / pitch diameter, pitch, included angle, helix angle, etc, must be known in order to carried out the design of the threaded joint.

DESIGN OF BOLTED JOINT

STRESS IN BOLTED JOINTS

For an eye bolt supporting a load acting along its axis and tending to stretch it or a hook bolt supporting a load twisting machine that is lifted by the threaded end of a cargo, in whichever case, each bolt is subjected to or carrying a tensile load P .

$$\text{Thus, } P = A d_e \sigma_t$$

(39)

Or the root diameter, $d_r = \sqrt{\frac{4P}{\pi \sigma_t}}$

Where P = Tensile load acting along the bolt axis, N
 σ_t = Allowable tensile stress of the material of the bolt, N/mm^2 .

Also, it is necessary to determine the stresses in screw fastenings due to bolt static and dynamic loading in order to determine their dimensions. In order to design for static loading both initial tightening and external loading need to be known.

INITIAL TIGHTENING LOAD

When a nut is tightened over a screw, the following stresses are induced:

- (i) Tensile stress due to stretching of the bolt.
- (ii) Torsional shear stress due to frictional resistance at the threads.
- (iii) Shear stress across threads.
- (iv) Compressive or crushing stress on the threads.
- (v) Bending stress if the surfaces under the bolt head or nut are not perfectly normal to the bolt axis.

TENSILE STRESS

Since none of the above-mentioned stresses can be accurately determined, bolts are usually designed on the basis of direct tensile stress with large factor of safety. The initial tension in the bolt may be estimated by an empirical formula or relation,

$$P_t \text{ (in N)} = 2480d, \text{ Where } d = \text{Nominal}$$

diameter of the bolt, mm.

The relation is used for marking the joint leak

Proof, such as the joint between the cover and cylinder of an internal combustion engine in which the nuts have metal to metal contact. However if leak proofing is not required, half of the estimated load is used, that is, P_i (in N) = 142000, where d_0 = Nominal diameter of the bolt, mm.

The reaction applies to the bolt fastening two parts with a flexible gasket between them. However, since initial stress is inversely proportional to the square of the diameter,

$$D_2 = \frac{2840 d_0}{4} = \frac{P_i}{\pi d_0^2} \Rightarrow d_0 = \sqrt{\frac{4 P_i}{\pi D_2}}$$

The expression indicates that screws are subjected to an excessive high initial tightening stress. Hence bolts that must carry load should not be under M16 or 16 mm. In such case, torque wrenches may be used to apply known load. Torque in wrench is given by,

$T = K P d$, Where K is a constant dependent on coefficient of friction at the mating surfaces. P = Tightening up load, N and d = bolt diameter, mm.

TORSIONAL SHEAR STRESS $\left[\tau = \frac{T}{J} \right]$

It is given by,

$$\tau = \frac{16 T}{\pi d^3} \quad \text{Where } T = \text{Torque due to tightening}$$

$$\tau = \frac{16 M_t}{\pi d^3} \quad \text{Where } M_t = \text{Torsional Moment applied}$$

$$N_{mm} = \tau \quad (\text{the Torque applied})$$

(41)

The normal stress in the lateral section of the threaded portion of the bolt, caused by tightening force, is given by,

$$D_t = \frac{P}{\pi d_c^2}, \text{ where } P = \text{tightening force, N,}$$

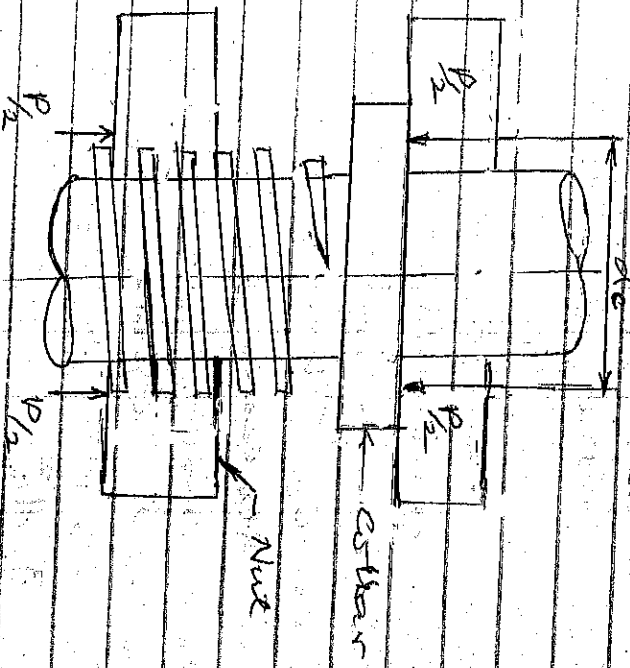
$$d_c = \text{core diameter of the thread, mm.}$$

$$= \sqrt{\frac{4P}{\pi}}$$

Now relating torque T to the tightening load P in a power screw configuration as shown below and taking collar friction into account yield,

$$T = \frac{P d_m (1 + \mu \pi d_m \sec \alpha)}{2} + \frac{P \mu_c d_m}{2}$$

Where d_m and d_m = the mean thread diameter and mean collar diameter respectively, and μ and μ_c = coefficients of thread and collar friction respectively, and α = semi thread angle. If $d_m = d_m + 1.5 d_m$ then $T = C P d_m$, where $C = \frac{1}{2} \sec^2 \alpha$ a given arrangement. The expression is similar to the one used to find the torque in a wrench.



Or alternately torque or torsional moment

can be determined by (4.2)
 T or $M_t = P \cdot d_m \tan(\alpha + \phi)$ (if forced)
 or $T = P \cdot d_m \tan(\alpha + \phi)$ (if collar)

Where P = Axial force of bolt tightening
 N , d_m = Pitch / mean diameter of the threads
 mm. α = Helix angle of the threads, °
 ϕ = Reduced angle of friction between

the surfaces of screw and nut, ° = $\tan^{-1} \mu$.
 μ = Coefficient of friction between the
 surfaces of screw and the nut.
 So when the bolt is subjected to

simultaneously to axial force and
 twisting moment / torque by the tight-
 ening of the bolt, the equivalent
 stress will be given as,

$$\sigma_{eq} = \sqrt{\sigma_t^2 + 4\tau^2} \quad \text{ie, Maximum Shear Stress theory,}$$

$$= \sqrt{\sigma_t^2 + 3\tau^2} \quad \text{ie, Maximum distortion energy theory.}$$

By considering the various factors of
 the joint such as magnitude of
 tightening force, parameters of the
 screw joint and the value of μ , it
 has been seen that for metric
 thread, the equivalent stress, on
 average is approximately 30% higher
 than stress σ_t . Thus, the effect of
 shearing stress can be considered
 for by increasing the design tensile
 load (initial tightening load) by 30%.

$$\therefore \sigma_t = 1.3 P \Rightarrow d_e = \sqrt{1.3 P}$$

$$\frac{\pi d_e^2}{4} \quad \frac{\pi d_t^2}{4}$$

(48) SHEAR STRESS ACROSS THREADS

It is given by,

$$I = 3P$$

Where d_c = Core diameter, mm.

$$\pi d_c b n$$

Where b = Base

Width of the thread.

Width of the thread section at

the root, mm.

$$n = \text{Number of threads}$$

engaging the lead / Number of threads in

engagement. P = Load force of bolt tightening

, N.

CRUSHING STRESS ON THREADS

It is given by,

$$D_c = P$$

$$\frac{\pi (d_o^2 - d_c^2)}{4} n$$

Where d_o and d_c =

Outer and Core dia -
Number of Thread
tightening, N.

BENDING STRESS

If the underside of the bolt and the bolt part are not parallel or shown below, the bolt may be subjected to bending and also if the outside surfaces of the parts there are

the diagram of the non-parallel are shown in, the bolt will be subjected to bending. Hence the bending stress is given by the equation

$$I_b = \frac{QE}{2L}$$

Where Q = Distance between the

of the nut or head and the bolt is under bending, or out of parallelism of one face with

respect to the other. L = Total thickness of

2 Unit Card

 $\lambda_1 \lambda_2$ 

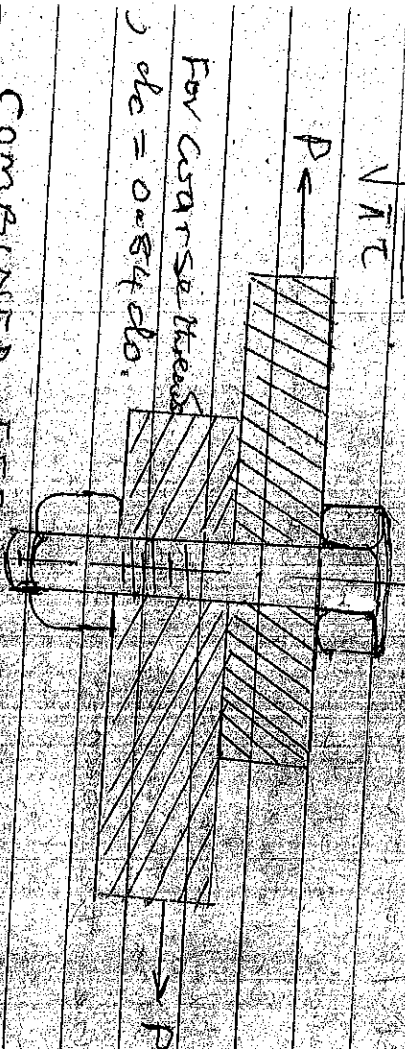
Hy Non-parallel flange

Bending moment $M/E = P \cdot d_0$

$$D_3 = M_3 = 0$$

$$h = \sqrt{\frac{8P}{A}}$$
$$Z = \frac{440}{\pi \cdot 0.002}$$

(45)
 diameter of the threads mm. $15 = \text{Number of threads}$, $P = \text{load applied}$, N
 $\therefore d_0 = \sqrt[4]{\frac{4KP}{\pi \tau}}$



For coarse threads
 $d_0 = 0.84 d_0$

COMBINED EFFECT OF INITIAL TIGHTENING LOAD AND EXTERNAL LOAD

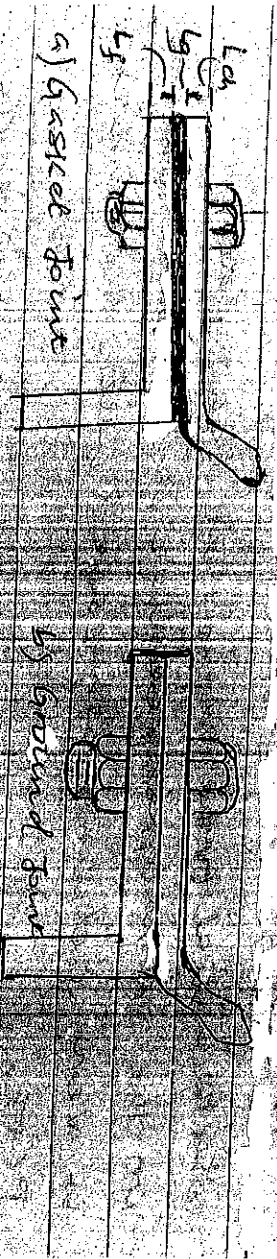
Consider bolted assembly shown in figure with an initial load P_0 is shown below in the figures a and b. The former shows a soft gasket placed between the parts and the latter shows metal to metal contact between the parts. In each case due to tightening, the bolt is elongated and the connected parts are compressed. If an external load, such as steam pressure is applied as tensile load on the bolts, the bolt elongation increases and the compression of the connected parts decreases and this depends on the relative elastic yielding of the bolts and connected parts.

As for the figure b, the resultant load will be either the initial load or the external load depending on the one that is greater. In case the bolt is very yielding compared with the connected part.

As for the figure a, the connected parts are very yielding compared with the bolt

and hence the resultant load on the bolt will be closely approximate the sum of the initial tension and the external load. Since the yielding of the bolt and the compressed parts involved depends on their stiffness, the amount of elongation of the bolt and that of the compressed parts will not be the same.

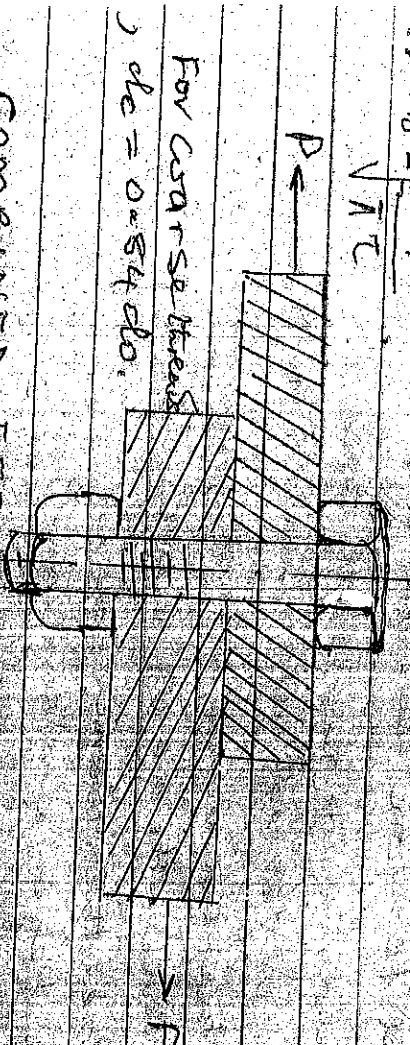
The tensile deformation of the bolt due to initial tightening is represented on the diagram below from left to right and that of the compressed parts from right to left on the x-axis of the diagram. The latter is bearing negative sign. The slope of the deformation curves of the bolt and compressed parts are also indicated. Their elongation diagram after the external load is applied is superimposed on the diagram next to the first one where the increase in extension of the bolt and the decrease in compression of the parts can be seen and so can be the increase in load on the bolt and the marking the decrease in compression of the parts.



(15)

diameter of the threads, mm. $15 = \text{Number of stress, } P = \text{load applied, N.}$

$$\therefore d_0 = \sqrt{\frac{4KP}{\pi \tau}}$$



COMBINED EFFECT OF INITIAL TIGHTENING LOAD AND EXTERNAL LOAD

Consider bolted assemblies each patterned with an initial load P_i as shown below in the figures a and b. The former shows a soft grade of

plate and the latter shows metal to metal contact between the parts.

In each case due to tightening, the bolt is elongated and the connected parts are compressed. Is an external

load, such as stress pattern, is applied as tensile load on the bolts, the bolt elongation increases and the compression of the connected parts decreases and this depends on the relative plastic yielding of the bolts and connected parts.

As for the figure b, the resultant load will be either the initial or the external load, depending on the one that is greater and also the bolt is very yielding compared with the connected part.

As for the figure a, the connected parts are very yielding compared with the bolt

(47)

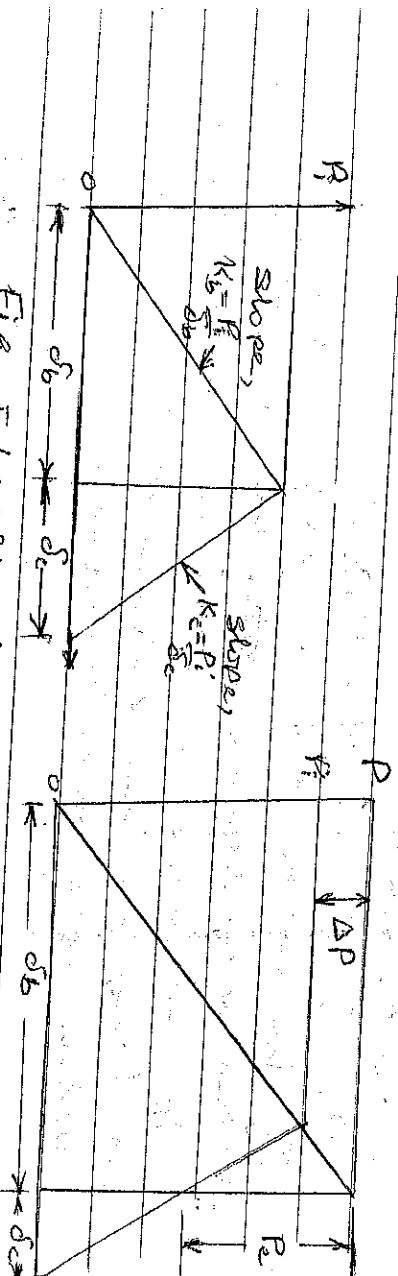


Fig. Elastic Behaviour of a bolted joint
The increased load P on the bolt is given by,

$\Delta P = k_P e$, where P_e = fraction of the external load, N .

k = Stiffness of the bolt to that of total stiffness.
= $\frac{1}{\frac{1}{k_b} + \frac{1}{k_n}}$, where q = Ratio of the elasticity of components

= k_b where k_b and k_n = Stiffness of the bolt and nut respectively, N/mm .

But $k_b = \frac{A_b E_b}{L_b}$, where L_b = Total length of parts held together by the bolts, i.e. cylinder head flange

flange and cylinder flange
= $L_{ch} + L_g + L_f$.

And $k_n = \frac{1}{\frac{1}{k_{en}} + \frac{1}{k_g} + \frac{1}{k_f}}$ A_b = Cross section area of the bolt

E_b = Modulus of elasticity of the bolt

L_b = Length of the bolt

It is assumed that area of flange is constant and bolt is a hollow cylinder of outer diameter, 3rd and inner diameter d .

$$\therefore A_e = \frac{\pi}{4} ((3d)^2 - d^2) = \frac{\pi d^2}{4} (9 - 1) = 2\pi d^2$$

$$\therefore k_b = \frac{2\pi d^2 E_b}{L_b} \quad k_g = \frac{2\pi d^2 E_g}{L_g}$$

$$k_f = \frac{2\pi d^2 E_f}{L_f}$$

$$\frac{1}{k} = \frac{1}{k_b} + \frac{1}{k_g} + \frac{1}{k_f}$$

(48)

Now the resultant load is given by,
 $P = P_i + \Delta P = P_i + K P_e$.

If the initial tension and the fraction of the external load used are known, the resultant load can be calculated as long as the value of K in the table below are substituted.

S/No	Type of joint	K
1.	Metals to metal joint through bolts.	0.02
2.	Hard Copper gasket with long through bolts.	
3.	Soft Copper gasket with long through bolts.	0.25
4.	Copper asbestos gasket.	0.50
5.	Soft packing with through bolts.	0.60
6.	Soft packing with studs.	0.95
7.	Normal Copper lining.	1.00
8.	Lead gasket with studs.	0.01
		0.10

Now the bolt designed on the basis of static failure can be mainly tightened up with initial tension greater than the external loads. This is the reason for practice in the assembly of automobile engines to tighten up connecting rod bolts so that the stress approaches the yield point. The one designed on the basis of cyclic failure can be tightened up with initial tension not higher enough, that is the resultant load on the bolt should be affected by the external loads.

Depending upon the resilient properties of the bolts and connecting parts, the resultant load acting on bolts will be within,

$$P = (1.1 \text{ to } 2) P_e$$

(50) N/mm^2 . $V = \text{Volume of the bolts}$, $mm^3 = A \times L$.

By this equation, the energy absorbed by each unit volume of the bolt material will be proportional to the square of the stress of the same material, hence larger part of the energy will be absorbed at the region of the threaded part, making it vulnerable to failure. In order to avoid that, a bolt of uniform strength is made.

Now if the shank is reduced in size until it has the same area as the rest of the screw, the stress in the shank will be increased, the deformation of the shank will be increased and consequently it occurs that the capacity of the bolts resist shock loading will be increased.

Also, the capacity of the bolt to resist the stress can also be increased by increasing the length of the bolt. Other methods of making bolts of uniform strength are as shown.

(i) By reducing the shank area to rest area through drilling hole as shown in figure (a) below so that

$$\pi d_h^2 = \pi (d_o^2 - d_c^2)$$

$$d_h = \sqrt{d_o^2 - d_c^2}$$

(But the hole must be good for long bolts.)

(ii) By turning down the diameter of the shank of the bolt to the same diameter as the threads or some times even less. The method is cheaper, but the bolt is weaker in bending and twisting. Refer to figure (b) below.

(iii) When a bolt is used to hold two or more members / parts together, it is better if the original length must be kept.

(49)

By virtue of the recurring nature of the loading, 2 can be adopted so that

$P = 2P_e = P_e + P_e$ i.e. $P_e = P_e$. This relation ensures a pressure tight joint between the clamped parts.

However, the initial tightening of the bolts causes tangential stresses. Hence

$P = P_e + 1.3P_e = 2.3P_e$. On substituting into $d_c = \sqrt{\frac{4P}{\pi d_e}}$, Hence

$$d_c = \sqrt{\frac{4 \times 2.3 P_e}{\pi d_e}} = \sqrt{\frac{9.2 P_e}{\pi d_e}} \quad \text{wheels}$$

DYNAMIC STRESSES IN SCREWS

Bores holding the bearing caps on the internal combustion engine and the pistons on a bridge and on passenger and freight cars are subjected to surging and twisting. The bolts are usually applied to ends and hence subjected to dynamic loads but also for the ability to absorb energy.

Hence the impact stress due to falling weight 'F' from a distance of h to strike a certain given bar,

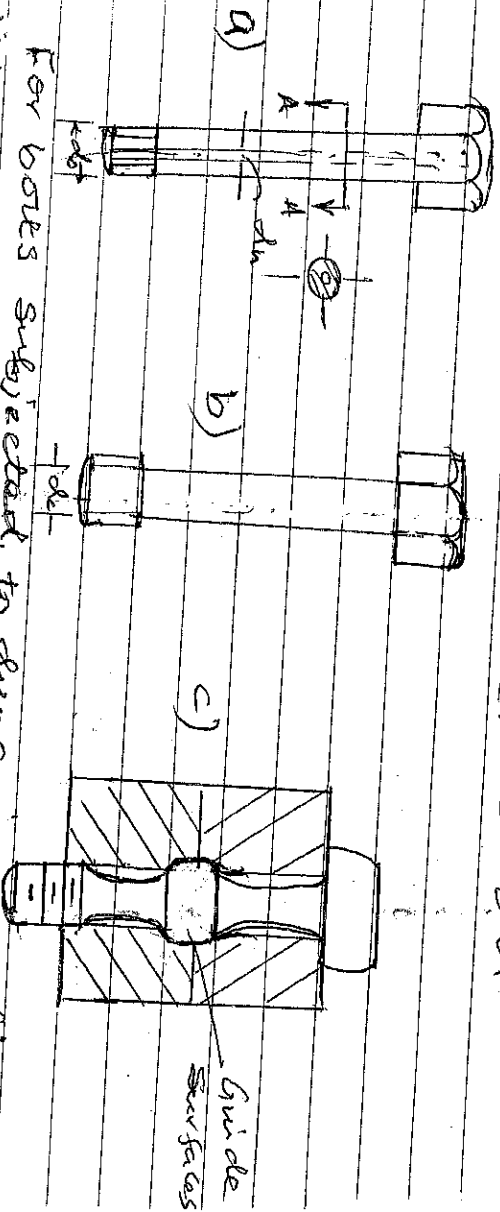
$$D_i = \frac{F}{A} + \frac{F}{A} \sqrt{1 + \frac{24h}{e}}$$

And the modulus of resilience R_e stored up in a cubic meter of material at the elastic limit and R_e is the

$$U_p = \frac{D_i P}{2E} \times V, \quad \text{Where } D_i P = \text{stress at the yield point}$$

$$U = \text{Modulus of Elasticity} \times \text{Volume} \times \text{Stress at the yield point}$$

each junction to serve as guide surface for the guided parts and for shear as shown in figure (a) below.



For bolts subjected to dynamic loading, the initial load is given as, $P_i = 2.5 \times \text{operating dynamic load}$ in axial direction.

DESIGN OF THE NUT

The thickness of the nut can be found from the strength of thread in shear, bending and compression.

Hence its thickness / height of the nut, $h_n = P \times n$, where $n = \text{Number of threads in engaged portion}$

And the shear stress is given by,

$$\tau \leq \frac{P}{\pi d h_n} \quad \text{where } P = \text{load, } N$$

the nut, mm.

It is worth noting that the tooth width at root may be equal to the thread pitch i.e. $b = P$.

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